

# Demonstration of Proposed Features

|                      |                                            |
|----------------------|--------------------------------------------|
| <b>Date</b>          | 5 July 2026                                |
| <b>Team ID</b>       | XXXXXX                                     |
| <b>Project Name</b>  | Voice-Based Concept Understanding Analyzer |
| <b>Maximum Marks</b> | 1 Mark                                     |

## Demonstration of Proposed Features

This document captures all the features proposed during the planning phase of the Voice-Based Concept Understanding Analyzer project and verifies whether each feature was successfully implemented and demonstrated. It provides evidence that the planned solution has been delivered successfully.

| S.No | Feature Name           | Description                                           | Status<br>(Implemented<br>/ Partial /<br>Pending) | Demonstrated<br>(Yes / No) | Remarks                               |
|------|------------------------|-------------------------------------------------------|---------------------------------------------------|----------------------------|---------------------------------------|
| 1    | Voice Upload           | Allows students to upload recorded explanations.      | Implemented                                       | Yes                        | Working successfully.                 |
| 2    | Speech-to-Text         | Converts uploaded speech into text using Whisper.     | Implemented                                       | Yes                        | Accurate transcription achieved.      |
| 3    | Semantic Analysis      | Compares student explanation with reference concepts. | Implemented                                       | Yes                        | Similarity score generated.           |
| 4    | Audio Feature Analysis | Extracts speech characteristics using Librosa.        | Implemented                                       | Yes                        | Waveform and audio metrics displayed. |

|   |                       |                                             |             |     |                                  |
|---|-----------------------|---------------------------------------------|-------------|-----|----------------------------------|
| 5 | Evaluation Dashboard  | Displays scores, strengths and suggestions. | Implemented | Yes | Interactive dashboard available. |
| 6 | PDF Report Generation | Generates downloadable performance reports. | Implemented | Yes | Reports generated successfully.  |

## Feature Implementation Summary

|                                 |      |
|---------------------------------|------|
| Total Features Proposed         | 6    |
| Total Features Implemented      | 6    |
| Total Features Demonstrated     | 6    |
| Overall Implementation Rate (%) | 100% |